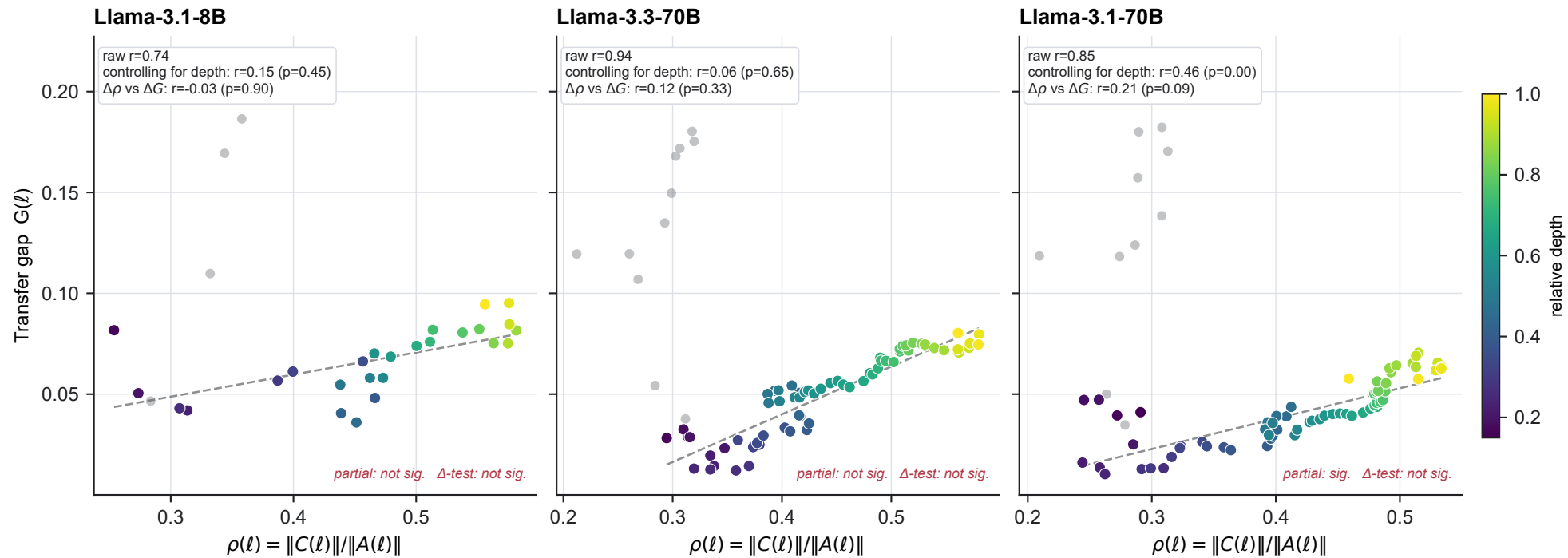


# Interaction geometry correlates with the transfer gap — almost entirely through depth



$\rho(\ell)$  tracks relative depth at  $r \approx 0.97-0.98$  in every model, and so does the gap — two depth-driven curves correlate with each other even absent a direct link. Controlling for depth (partial Spearman), the association is NOT significant in 2 of 3 models and survives only partially in the third (3.1-70B,  $r=0.46$  vs. raw 0.85). The harsher test — does a layer-to-layer increase in  $\rho$  coincide with a layer-to-layer increase in the gap ( $\Delta\rho$  vs.  $\Delta G$ ) — is non-significant in all three models. Read: most of the raw correlation is a shared-depth-trend artifact, not evidence that interaction geometry specifically drives the transfer-gap regrowth. Not promoted to a mechanistic claim.